

2-Mark Questions & Answers

Q1. Define instruction format.

An Instruction Format defines the internal layout and binary structure of a machine instruction. It specifies the bit fields allocated for the Opcode (operation code) and Operands (registers, memory addresses, or immediate values).

Q2. What is a zero-address instruction format? Give an example.

A Zero-Address Instruction Format contains only an Opcode and specifies no explicit address fields. It operates implicitly on a Stack data structure where operands are pushed/popped from the top of stack (`TOS`).

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Q3. Define control unit. What are its functions?

The Control Unit (CU) is the engine of the CPU that coordinates all computer hardware operations.

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Q4. What is the difference between hardwired and microprogrammed control unit? (any 2 points)

| Feature | Hardwired Control Unit | Microprogrammed Control Unit |

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Q5. Define microinstruction.

A Microinstruction is a low-level control word stored in Control Memory (ROM) that specifies one or more primitive hardware micro-operations (e.g., register transfer, ALU operation) to be executed within a single clock cycle.

Q6. What is a pipeline hazard?

A Pipeline Hazard is a situation or conflict in an instruction pipeline that prevents the next instruction from executing in its designated clock cycle, introducing pipeline stalls (bubbles) and reducing instruction throughput.

Q7. Define Flynn's classification. Name its four types.

Flynn's Classification is a taxonomy for categorizing computer architectures based on the number of concurrent Instruction Streams and Data Streams.

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Q8. What is instruction pipelining?

Instruction Pipelining is an architectural technique where the execution of sequential instructions is overlapped in time across multiple independent processing stages (e.g., Fetch, Decode, Execute), increasing instruction throughput.

Q9. Define structural hazard.

A Structural Hazard (resource conflict) occurs in a pipeline when two or more overlapping instructions require access to the exact same physical hardware resource (e.g., single memory bus or ALU) in the same clock cycle.

Q10. What is branch prediction?

Branch Prediction is a technique used by modern processors to predict the outcome of a conditional branch instruction (taken or not taken) before it is actually evaluated, allowing the pipeline to prefetch instructions without stalling.

Q11. State the functions of PC, IR, MAR, and MDR.

* **PC (Program Counter):** Holds the memory address of the next instruction to be fetched.

* **IR

Q12. What is RAW hazard?

A RAW (Read-After-Write) Hazard (also called True Data Dependency) occurs when an instruction tries to read a register or memory location before a preceding instruction has finished writing its updated result to it.
